SECTION 10

DESIGN AND SIMULATION OF A DIGITAL DATA TRANSMISSION ERROR DETECTION SYSTEM USING XOR LOGIC

PROBLEM STATEMENT

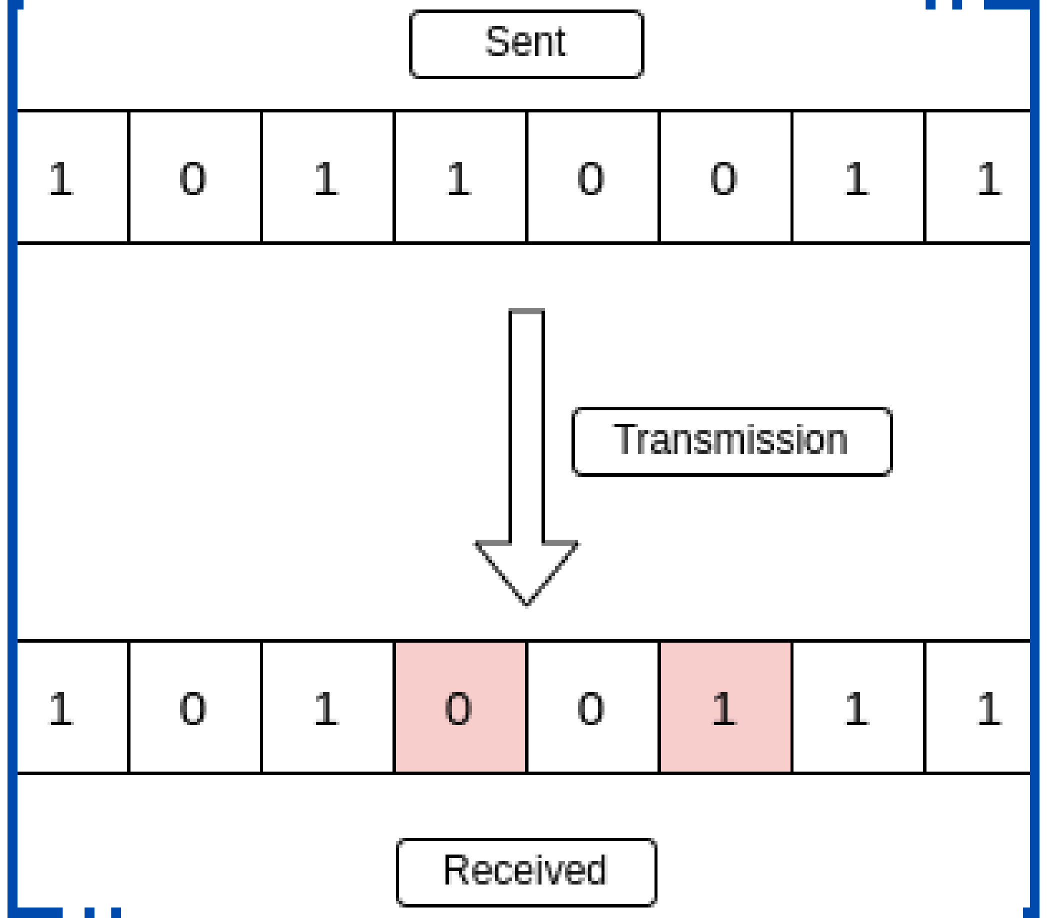
Transmission errors and ways to detect

Digital data can be corrupted during transmission due to noise, interference, or hardware faults. Even one changed bit can make the received data different from the transmitted data.

The problem is to design a simple system that can detect whether 4–bit data has been corrupted during transmission.

Proposed solution:
Use XOR gates and parity checking to detect transmission errors.

Example:
Transmitted: 1011
Received: 1001
→ Error should be detected.



DESIGN / METHODOLOGY

- 4-bit data is applied to the transmitter.
- XOR gates generate the parity bit.
- Data + parity bit are transmitted.
- The receiver checks the received bits using XOR logic.
- Output 0 → No error detected.
- Output 1 → Error detected.

For even parity:

$P = D3 \oplus D2 \oplus D1 \oplus D0$

4-BIT DATA

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XOR PARITY GENERATOR

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DATA + PARITY BIT

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TRANSMISSION

↓

XOR PARITY CHECKER

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ERROR / NO ERROR

DEMO IN LOGISIM + GITHUB REPOSITORY

LOGISIM RESULT

Correct data → Output 0 → No Error

One–bit error → Output 1 → Error Detected

CASE 1: CORRECT TRANSMISSION

Data = 1011, Parity = 1

Checker output = 0 → NO ERROR

CASE 2: ONE-BIT ERROR

One data bit is changed while the original parity is retained.

Checker output = 1 → ERROR DETECTED

GitHub Repository:

https://github.com/vpaakarapu-crypto/DESIGN-AND-SIMULATION-OF-A-DIGITAL-DATA-TRANSMISSION-ERROR-DETECTIONL-SYSTEM-USING-XOR-LOGIC

The Logisim simulation demonstrates that XOR-based parity checking can detect a single-bit transmission error.